Name: Thirishaanth S Register Number: 250701790

Experiment Name: EDA - Quantitative and Qualitative Analysis

```
import seaborn as sns
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline

tips = sns.load_dataset('tips')

print(tips.head())

sns.displot(tips['total_bill'], kde=True)
plt.show()

sns.displot(tips['total_bill'], kde=False)
plt.show()

sns.jointplot(x='tip', y='total_bill', data=tips)
plt.show()

sns.jointplot(x='tip', y='total_bill', data=tips, kind='reg')
plt.show()

sns.jointplot(x='tip', y='total_bill', data=tips, kind='hex')
plt.show()

sns.pairplot(tips)
plt.show()

print(tips['time'].value_counts())

sns.pairplot(tips, hue='time')
plt.show()

sns.pairplot(tips, hue='day')
plt.show()

sns.heatmap(tips.corr(numeric_only=True), annot=True)
plt.show()

sns.boxplot(y=tips['total_bill'])
plt.show()

sns.boxplot(y=tips['tip'])
plt.show()

sns.countplot(y=tips['day'])
plt.show()
```

```
sns.countplot(y=tips['sex'])
plt.show()

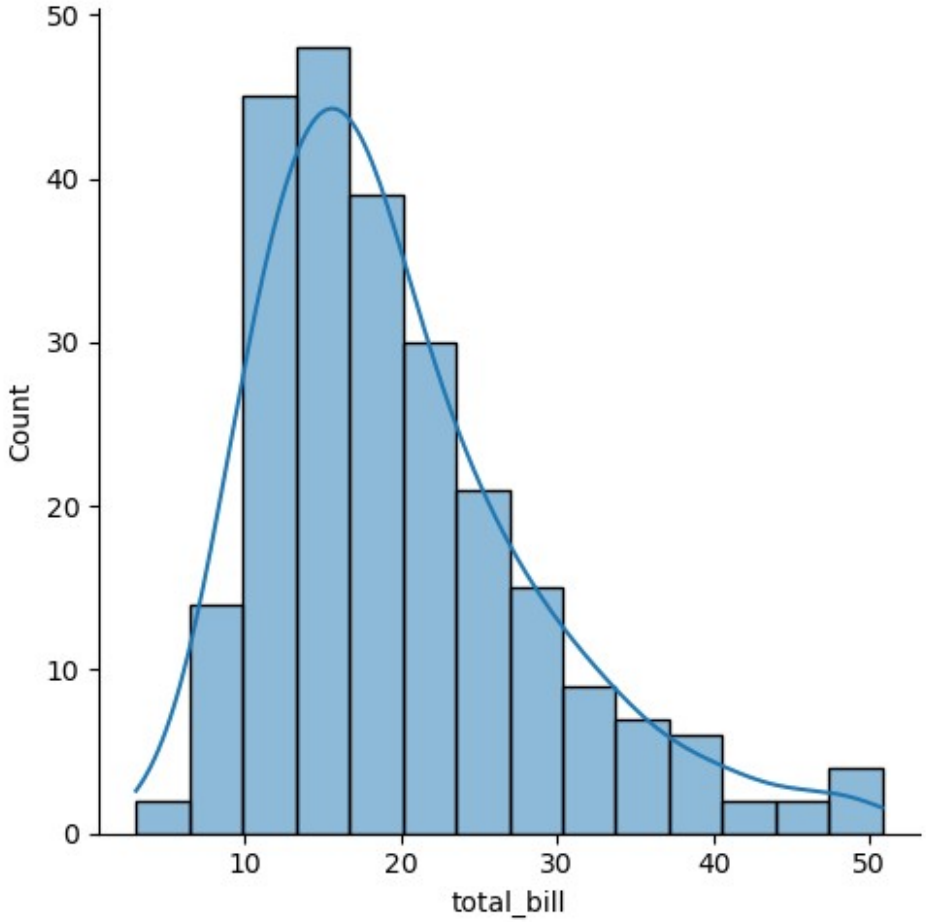
tips['sex'].value_counts().plot(kind='pie')
plt.show()

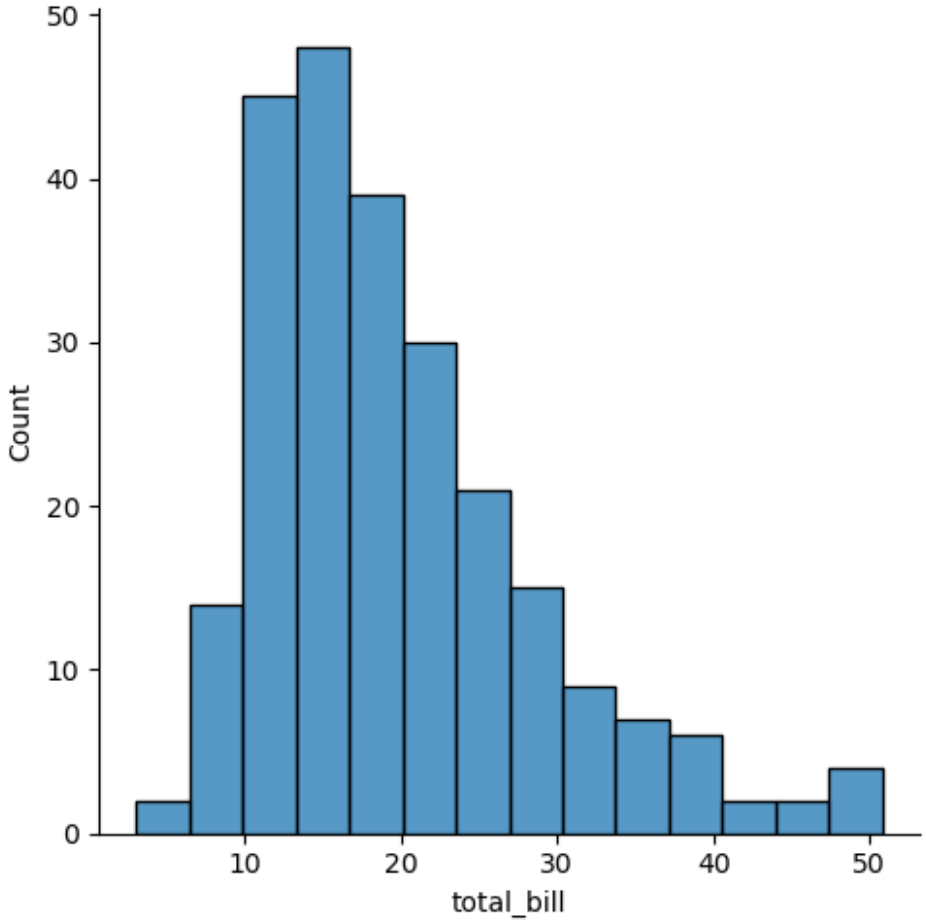
tips['sex'].value_counts().plot(kind='bar')
plt.show()

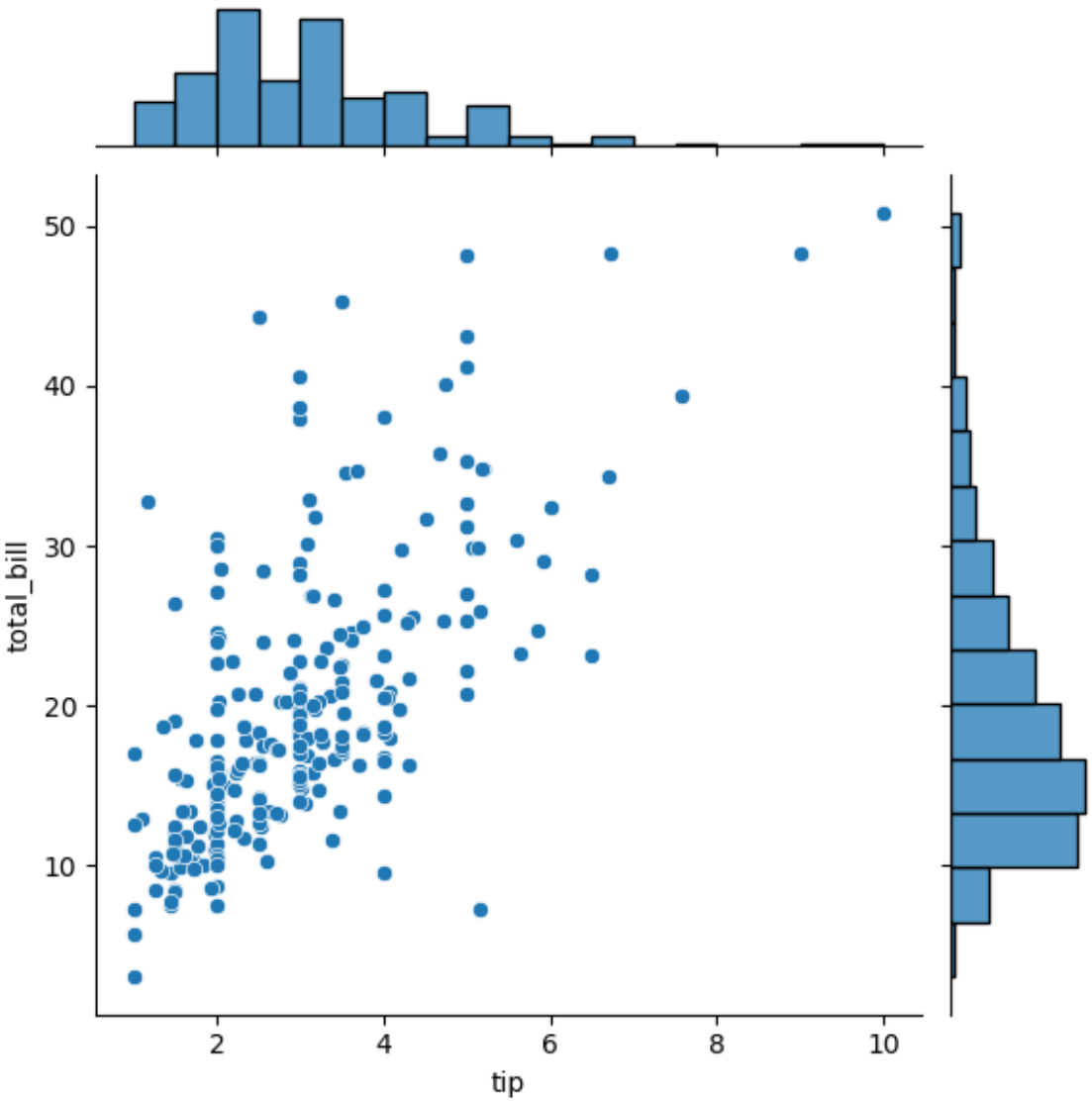
sns.countplot(x=tips['sex'])
plt.show()

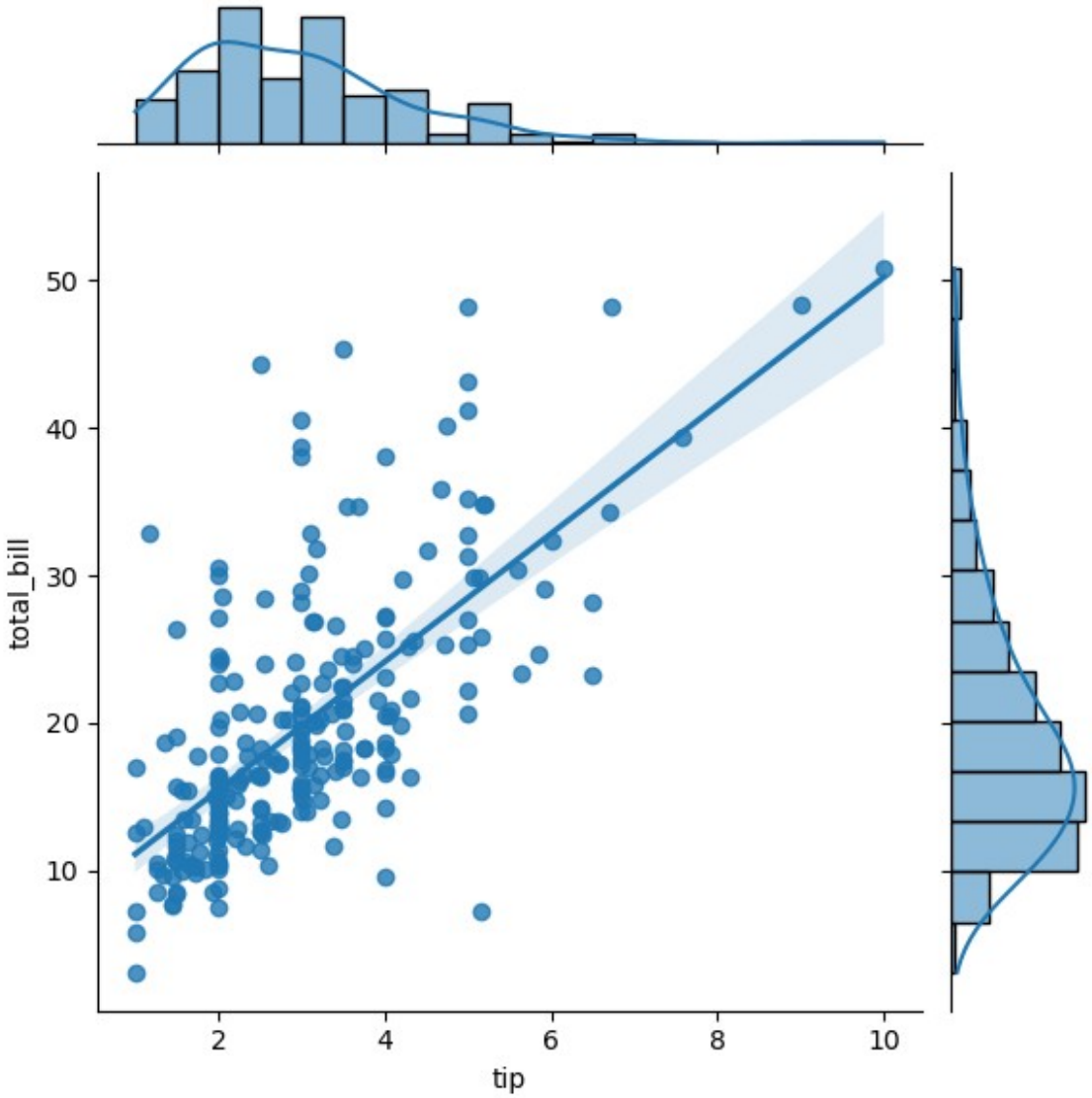
sns.countplot(data=tips[tips['time'] == 'Dinner'], x='day')
plt.show()
```

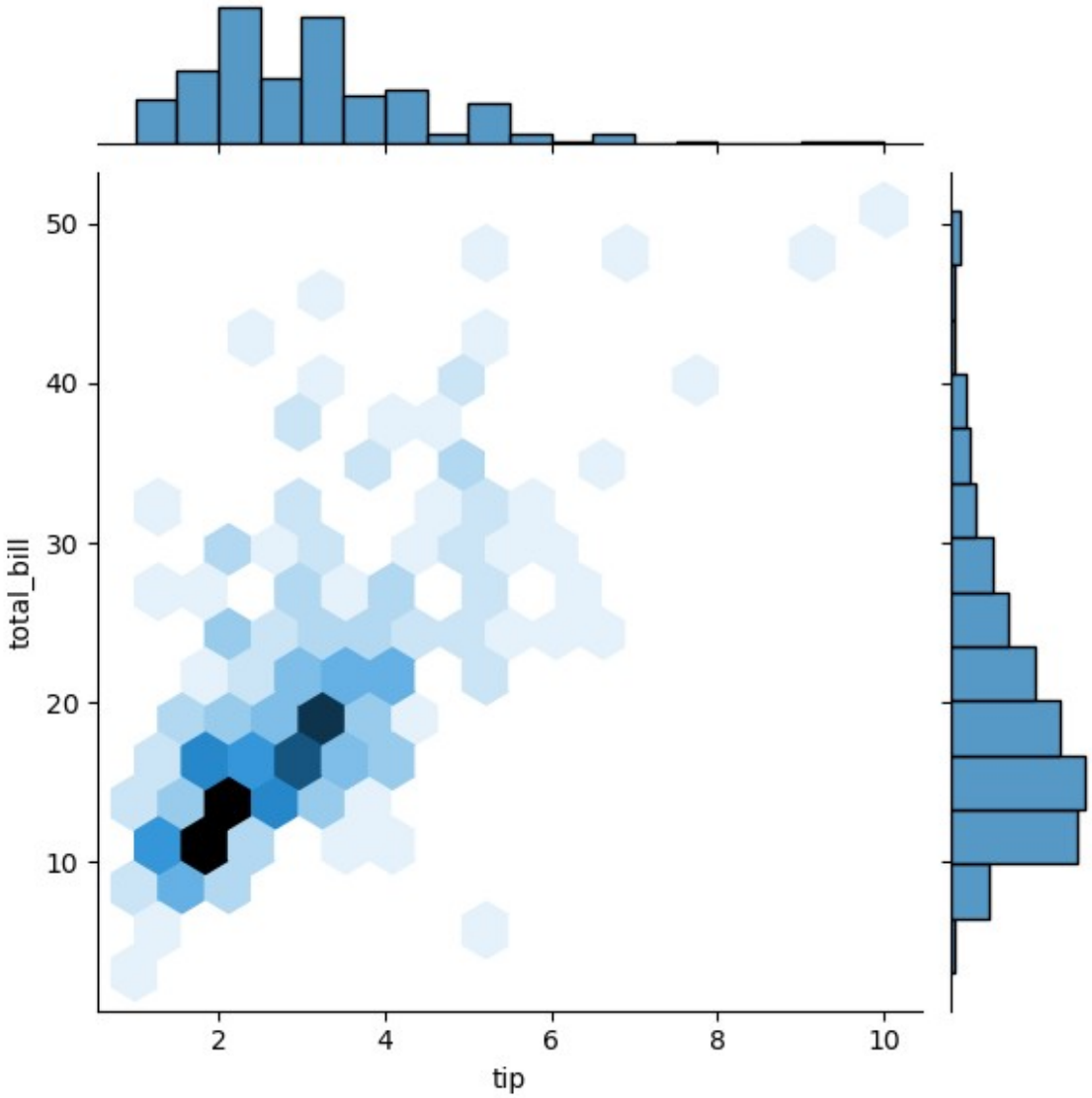
	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

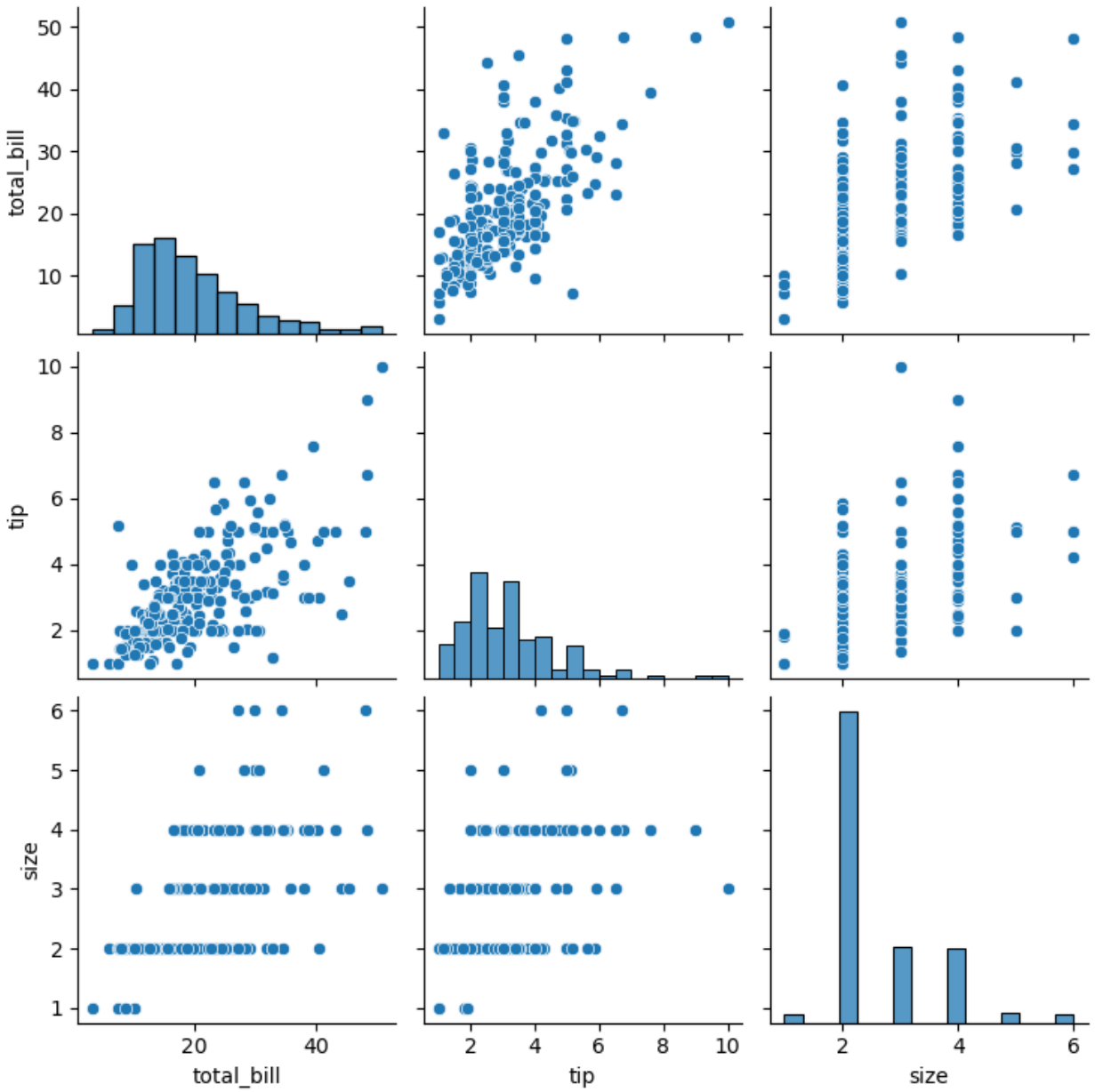

50
40
30
20
10
0
Count
10
20
30
40
50
total_bill


50
40
30
20
10
0
Count
10
20
30
40
50
total_bill

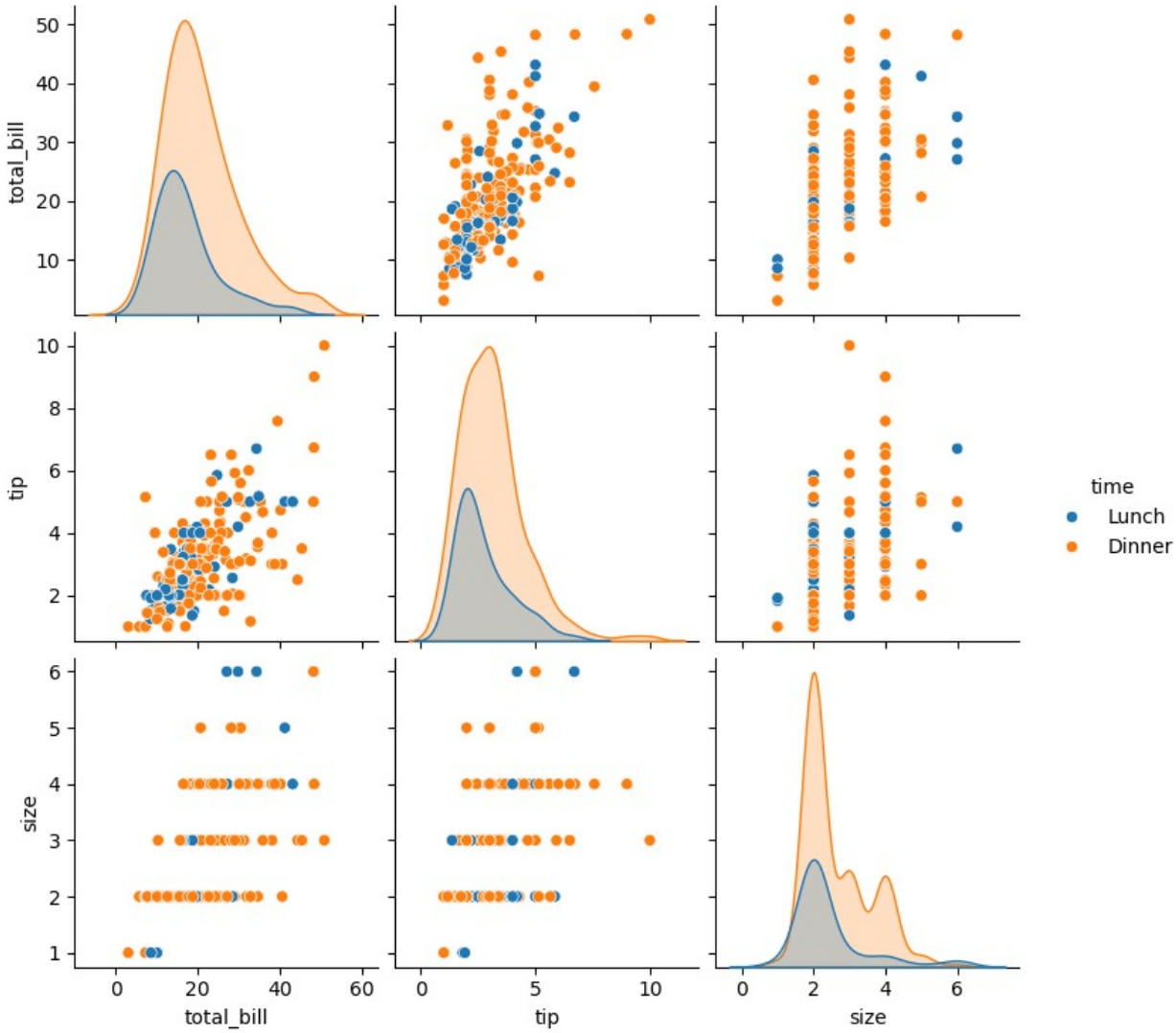

total_bill
tip
10
20
30
40
50
2
4
6
8
10

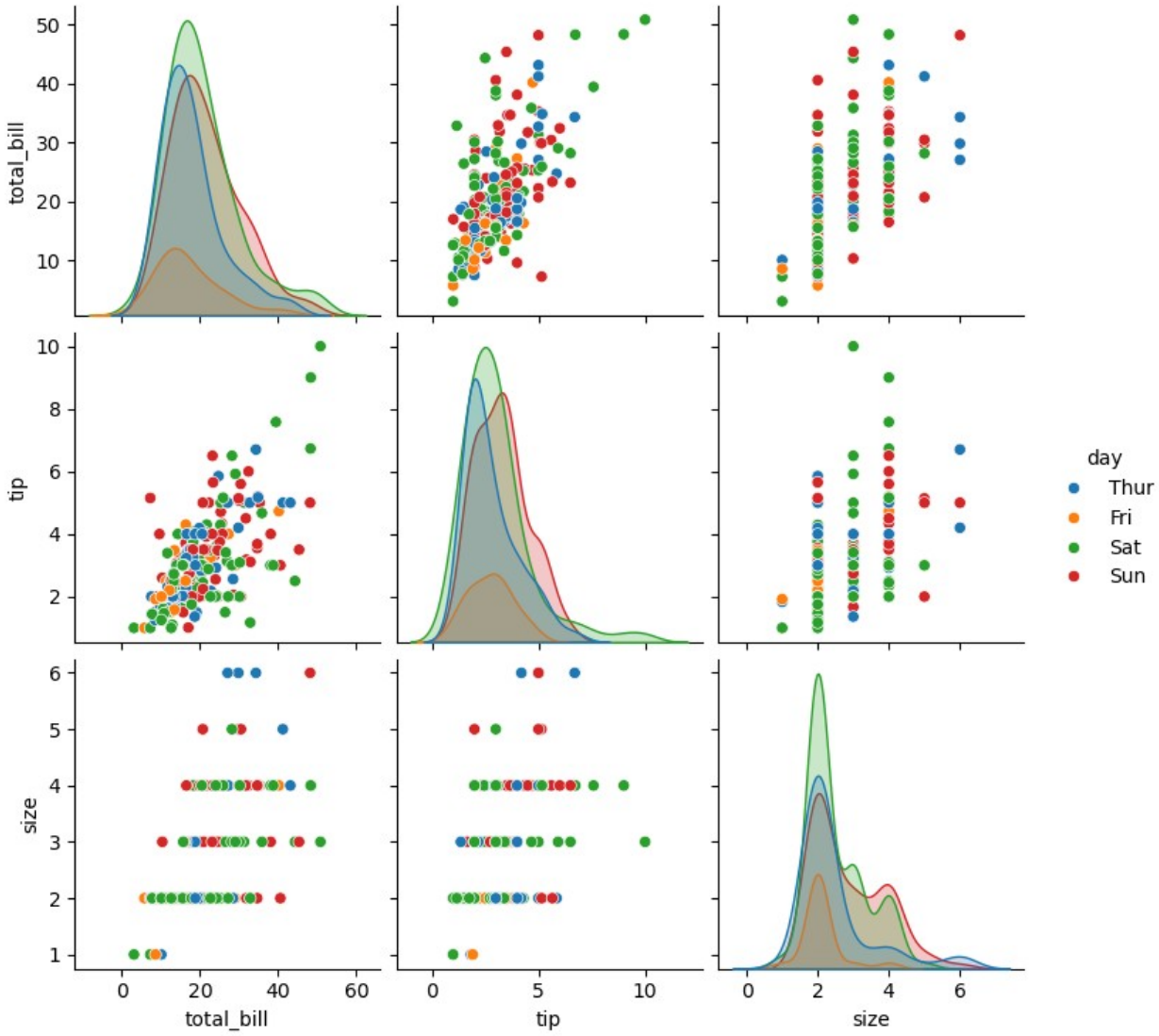

total_bill
tip
10
20
30
40
50
2
4
6
8
10

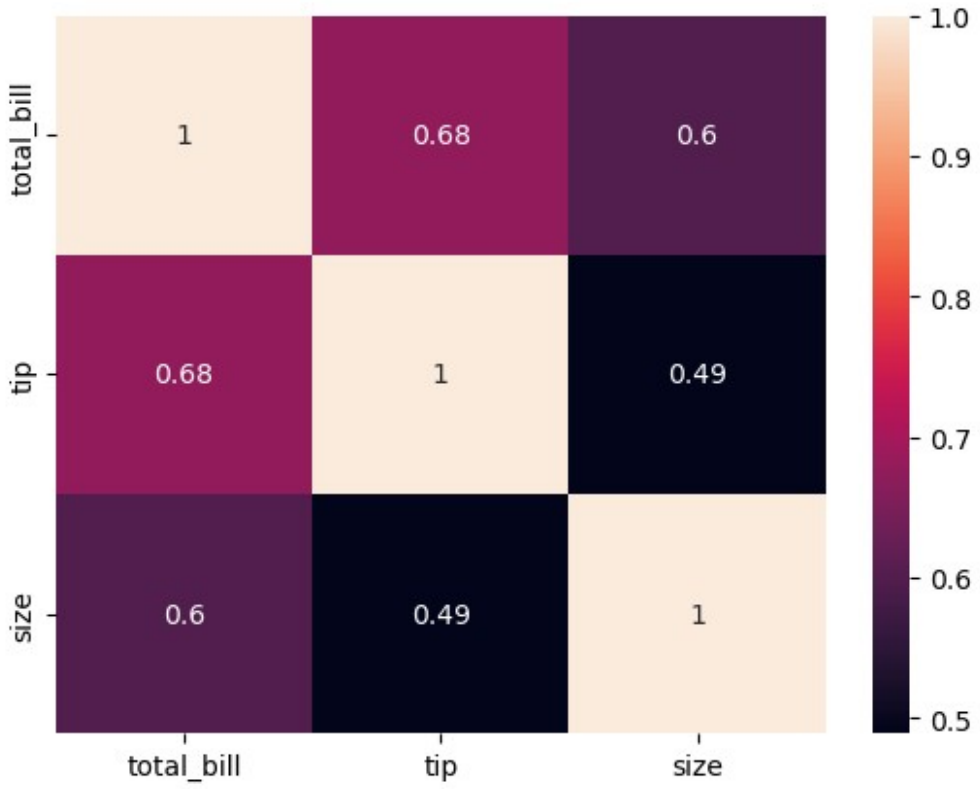

50
40
30
20
10
total_bill
2
4
6
8
10
tip

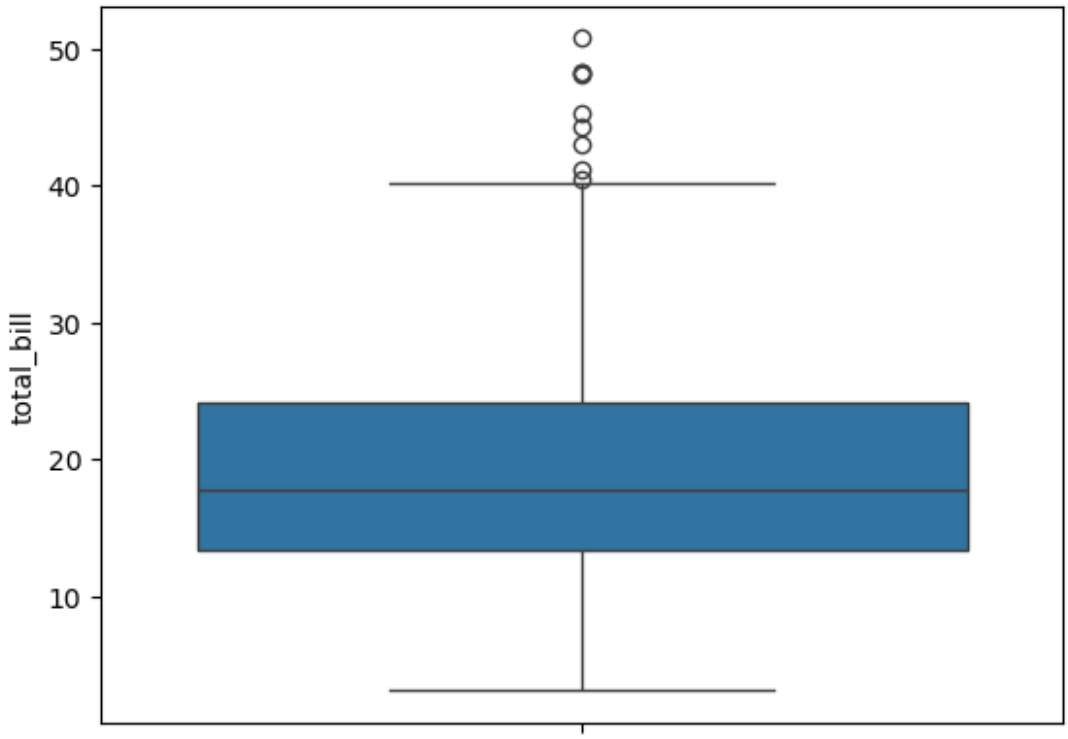


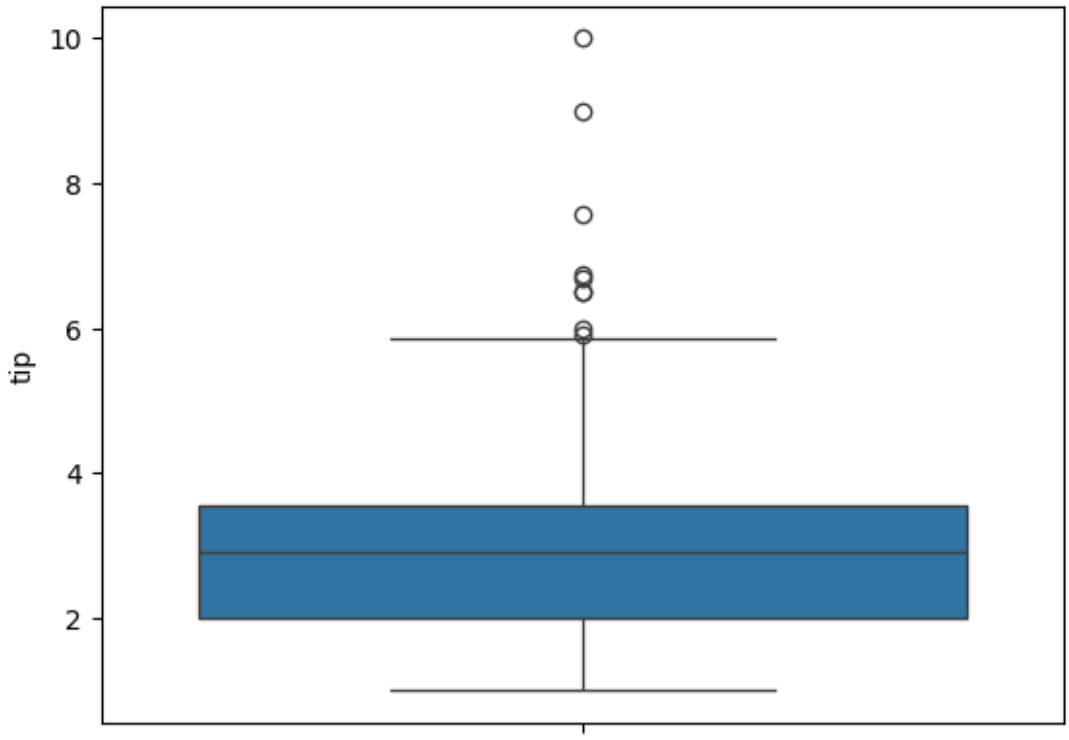

```
time
Dinner    176
Lunch      68
Name: count, dtype: int64
```

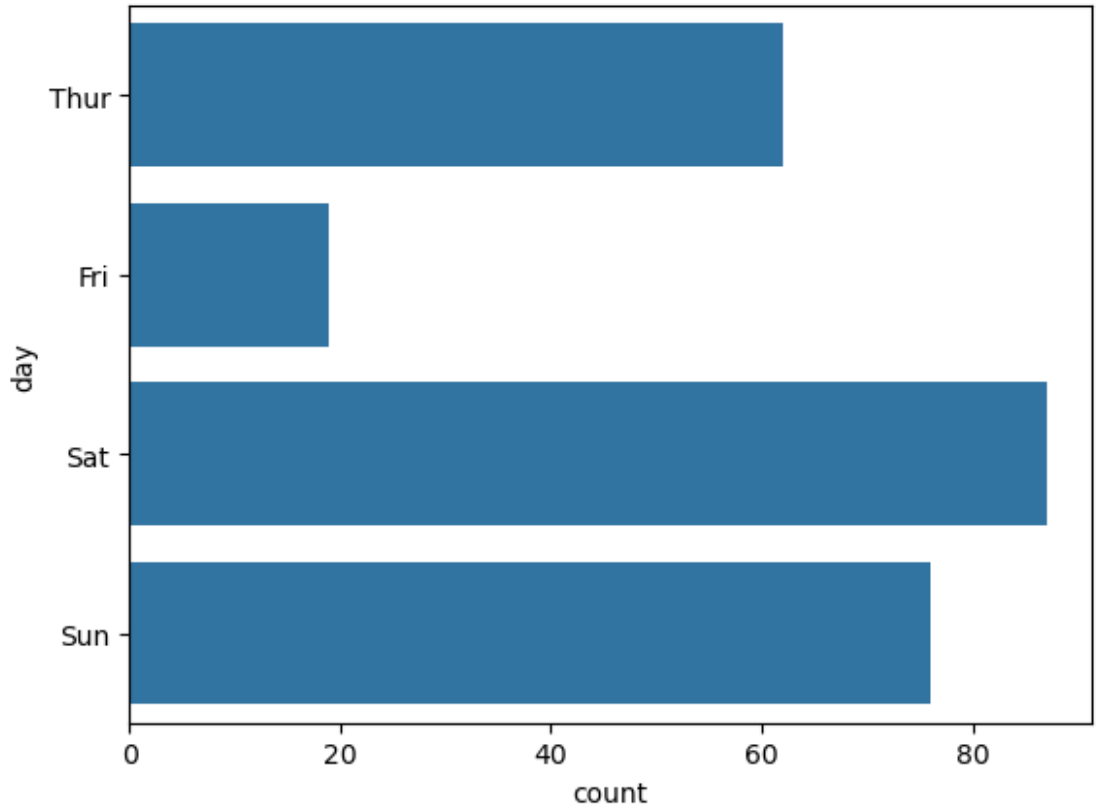

total_bill
tip
size
time
Lunch
Dinner

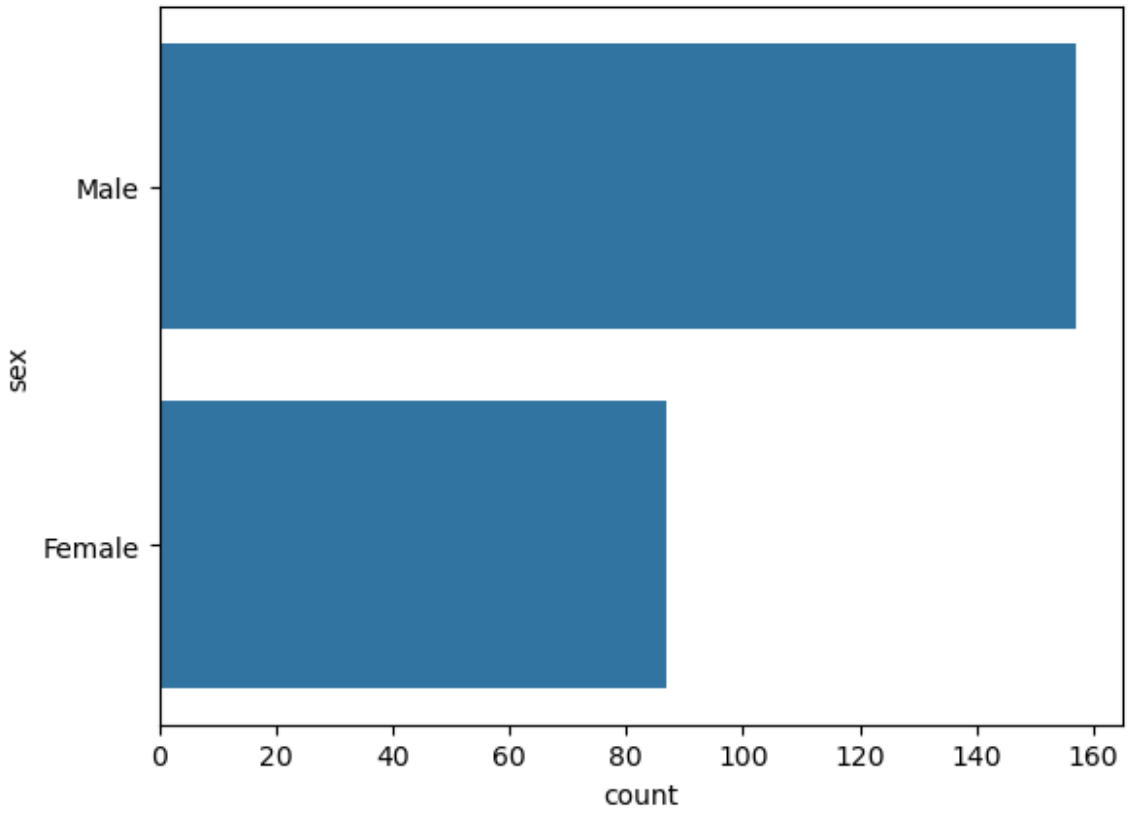

total_bill
tip
size
day
Thur
Fri
Sat
Sun

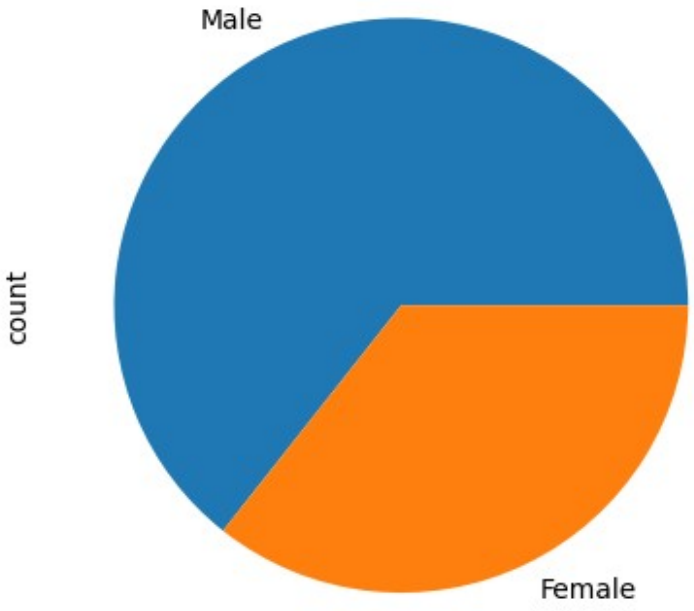

total_bill
tip
size
1
0.68
0.6
0.68
1
0.49
0.6
0.49
1
total_bill
tip
size
1.0
0.9
0.8
0.7
0.6
0.5

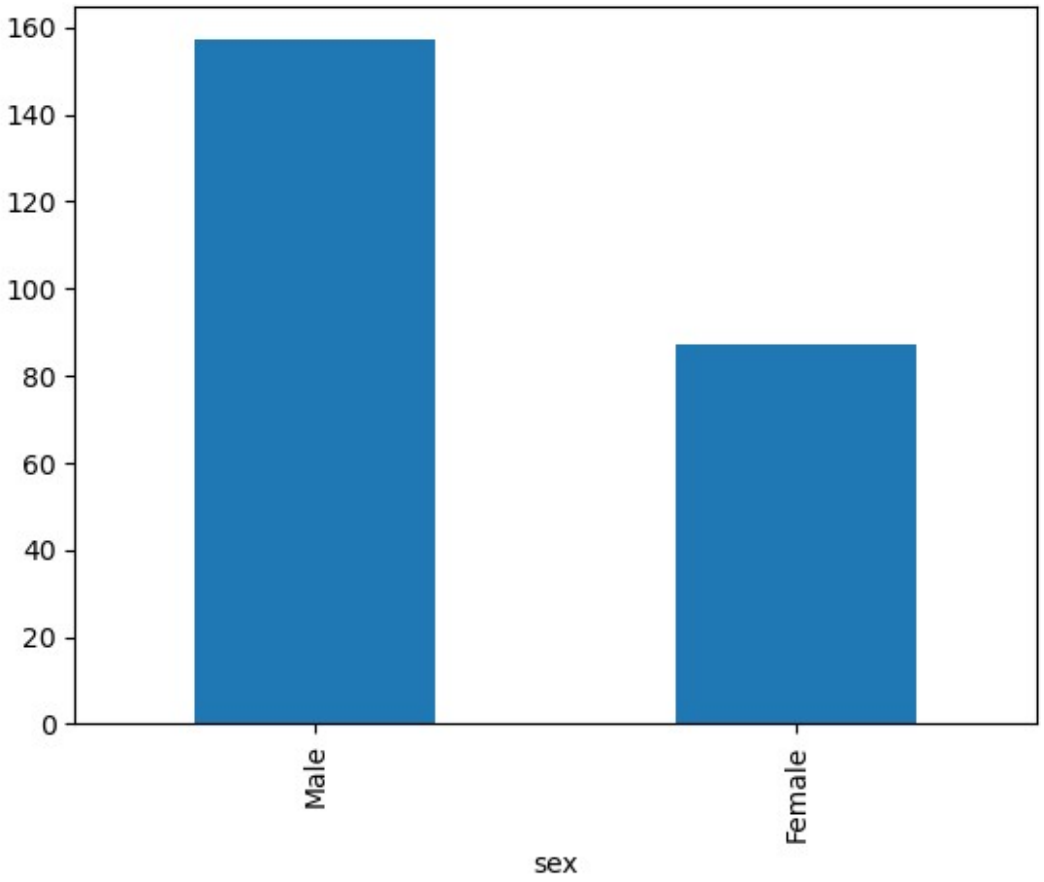

total_bill
50
40
30
20
10

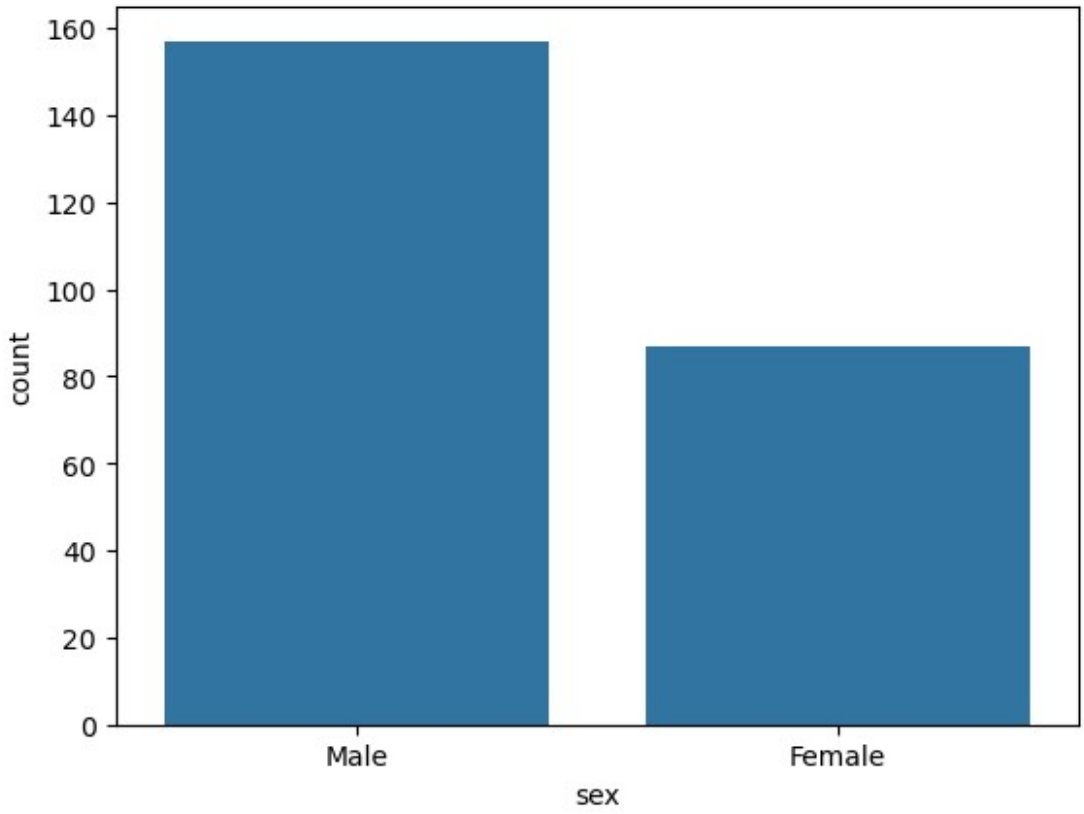

10
8
6
4
2
tip

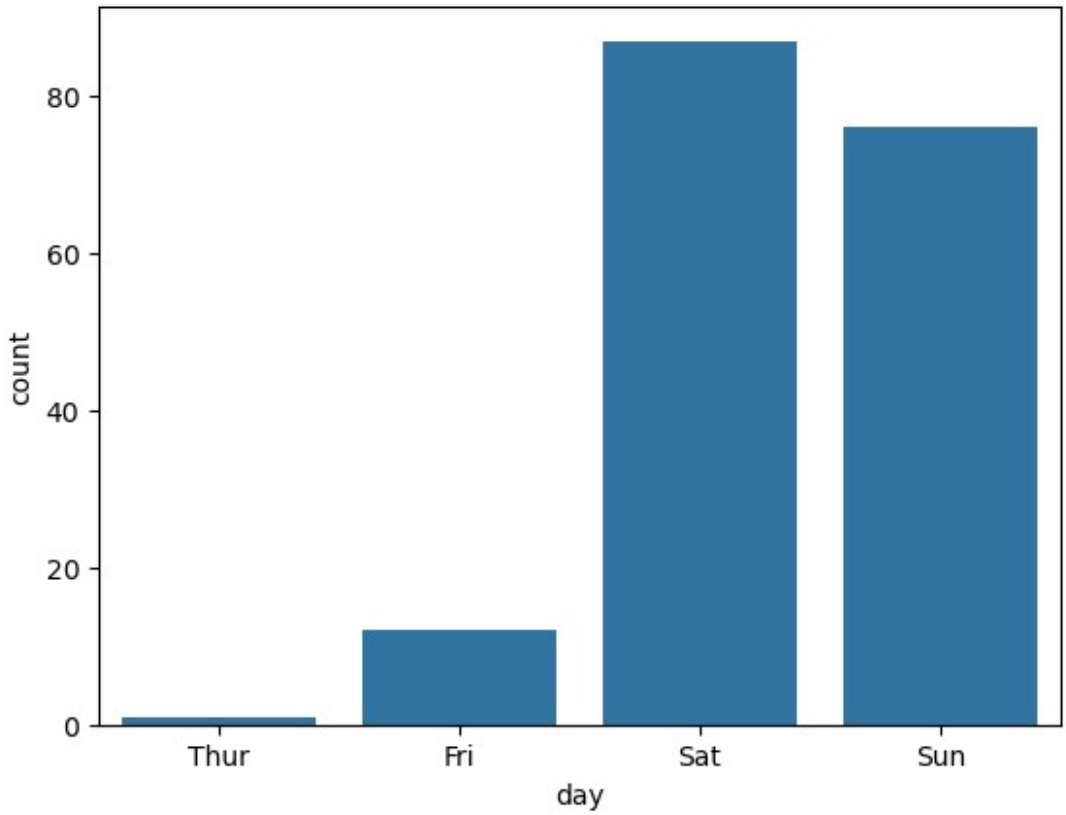

Thur
Fri
Sat
Sun
day
0
20
40
60
80
count


Male
Female
sex
0
20
40
60
80
100
120
140
160
count


Male
Female
count


160
140
120
100
80
60
40
20
0
Male
Female
sex


160
140
120
100
80
60
40
20
0
count
Male
Female
sex


80
60
40
20
0
count
Thur
Fri
Sat
Sun
day